Esercizio Investimenti – Società Lamin

LAMIN operates in the electronic devices manufacturing industry. Its flagship product is the DIH calculator, which is sold at a price of €1,000 per unit, with avoidable (variable) production costs equal to €500 per unit. The production workforce consists of three employees, each earning €25,000 per year
In 2019, the CEO evaluates a new investment opportunity. LAMIN could purchase a new machine to produce a next-generation calculator, called PAT, which would be sold at €1,300 per unit, with avoidable costs of €600 per unit.
Sales forecast for the four-year period 2020-2023, reported below. It should be noted that the introduction of PAT reduces the demand (and therefore production) of DIH.

	2020	**2021**	**2022**	**2023**
Demand PAT	1.500	1.600	1.700	1.800
Reduction Demand DIH	1.700	1.800	1.900	2.000

To do this, Lamin purchase price of the new machine: €400,000, expected life of 4 years. Acquisition date 1/1/2020, and the machine becomes effective immediately. Residual value at 2023: € 50.000, and the machine is sold. When evaluating the investment, the following effects must also be considered:

1. In addition to the 3 employees, an additional worker must be hired, with an annual salary of €35,000.;
2. Training costs of €20,000 per year will be incurred in 2020 and 2021.;
3. Additional administrative costs: € 10.000 (2020), € 12.000 (2021), and € 15.000 (2022, 2023);
4. The company obtains a subsidized loan from the Ministry of Industry for €100,000, at an interest rate of 5%, starting January 1, 2020. Yearly financial expenses related to this loan must be considered where appropriate.

*Compute the Net Present Value (NPV) of the investment after taxes, using the **invested capital** (Free Cash Flow to the Firm) approach.*

*Compute the Net Present Value (NPV) of the investment using the **equity** (Free Cash Flow to Equity) approach.*